Hold Out Test Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Hold Out Test Data before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Train a Regression Model

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Mean, Median, and Baselines

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Precision and Recall

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Hold Out Test Data trace and record the first step you can explain without help.

Use W2: Compare a baseline for Final testing and honest **claim**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Hold Out Test Data explanation against the supplied facts before discussing it.

Use the worked case for Final testing and honest **claim**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

A new frozen test has TN 7 FP 1 FN 2 TP 2. Calculate accuracy, precision and recall and state a bounded **claim**.

> **My independent explanation, trace and limits**
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> | |
> |---|
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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Hold Out Test Data target to a trace and a justified conclusion.

After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Hold Out Test Data case and show the deciding evidence.

A final-test folder has twenty leaves; its labels are opened weekly to choose models.

Is it still untouched?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

You inspect final test failures, change features to fix them and report the new score on that same test set as untouched. What is wrong?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Hold Out Test Data adds to Responsible Plant Classifier and keep the evidence.

Seal the Plant Classifier final test IDs and labels; record who may open them and at which release gate.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Calculate a final report

> Calculate a final report. Apply the supplied Hold Out Test Data facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Hold Out Test Data", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Calculate the worked accuracy, precision and recall with denominators.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Compare a baseline

> Compare a baseline. Apply the supplied Hold Out Test Data facts and keep your reasoning.

Why does 75 percent versus 70 percent not settle deployment?

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Handle a revision

> Handle a revision. Apply the supplied Hold Out Test Data facts and keep your reasoning.

The team changes a threshold after reading test errors. How should it label later results?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Hold Out Test Data facts and keep your reasoning.

A new frozen test has TN 7 FP 1 FN 2 TP 2. Calculate accuracy, precision and recall and state a bounded **claim**.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Report a failure

> Report a failure. Apply the supplied Hold Out Test Data facts and keep your reasoning.

One planned case returns no **prediction**. What must the report retain?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

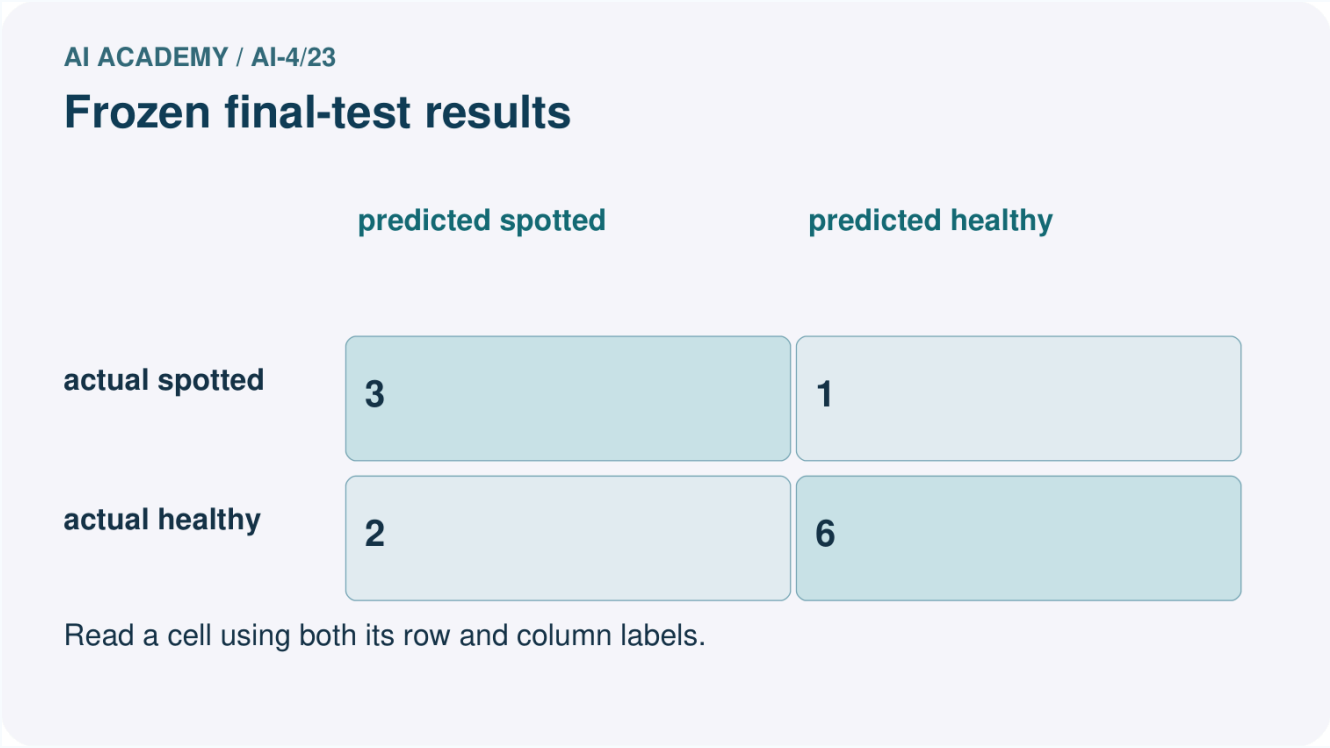
Pictorial case practice: Hold Out Test Data

> Open the final hold-out only after recording the frozen model choice. Explain the labelled step before checking the changed condition.

Open the final hold-out only after recording the frozen model choice.

Frozen plant classifier evaluates12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is **claim**ed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/23

Hold Out Test Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record freeze

Record model version and choices before reading these final references.



2. Calculate metrics

Accuracy9/12=75%; precision3/5=60%; recall3/4=75%.



3. Bound claim

Report these twelve held-out cases and their collection conditions; preserve all three error records.

CHANGE ONE THING

The team changes a threshold after inspecting the false negatives.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Open the final hold-out only after recording the frozen model choice.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report these twelve held-out cases and their collection conditions; preserve all three error records.

1. Record **freeze**

2. Bound **claim**

3. Calculate metrics

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Bound **claim** Report these twelve held-out cases and their collection conditions; preserve all three error records. Read the picture in order.

Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The team changes a threshold after inspecting the false negatives. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This set is now development **evidence** for the revised model.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: This set is now development **evidence** for the revised model. Using final outcomes to change the model breaks the untouched hold-out role.

Repair a missing building block

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

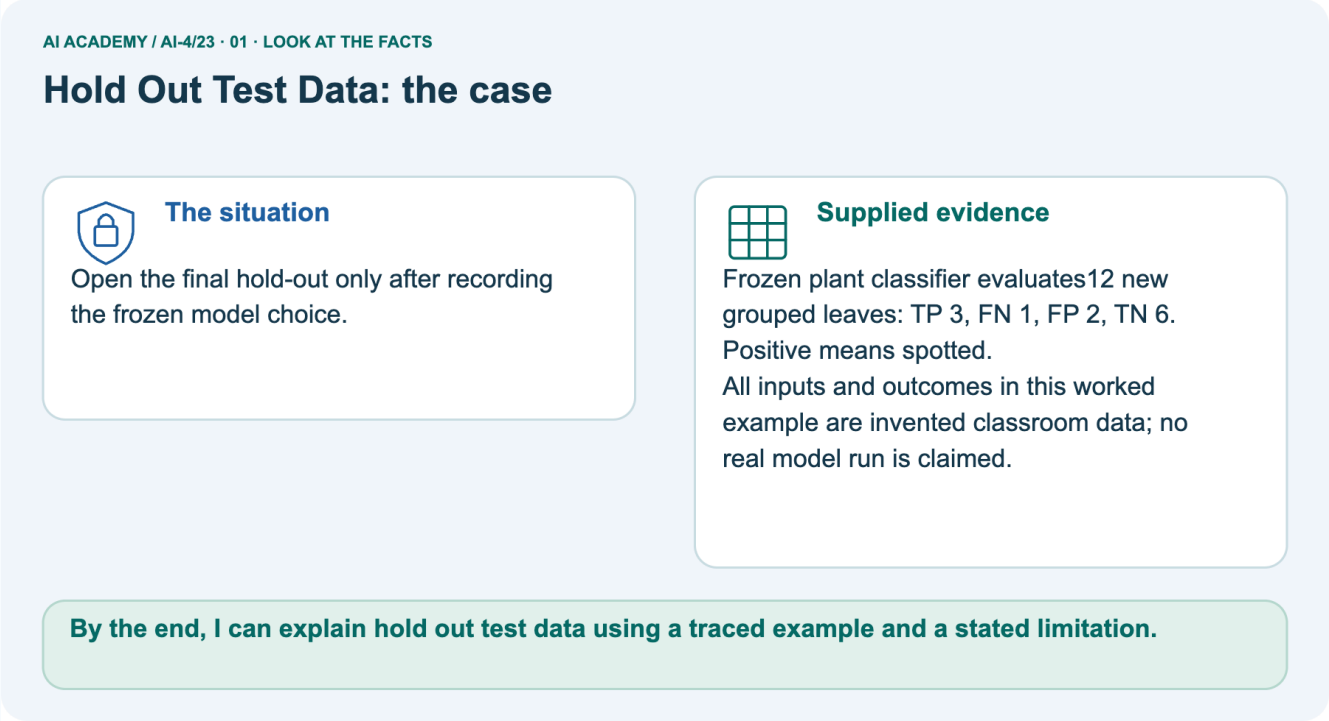
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Hold Out Test Data picture from its labels, then record a relevant limit.

01. Hold Out Test Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

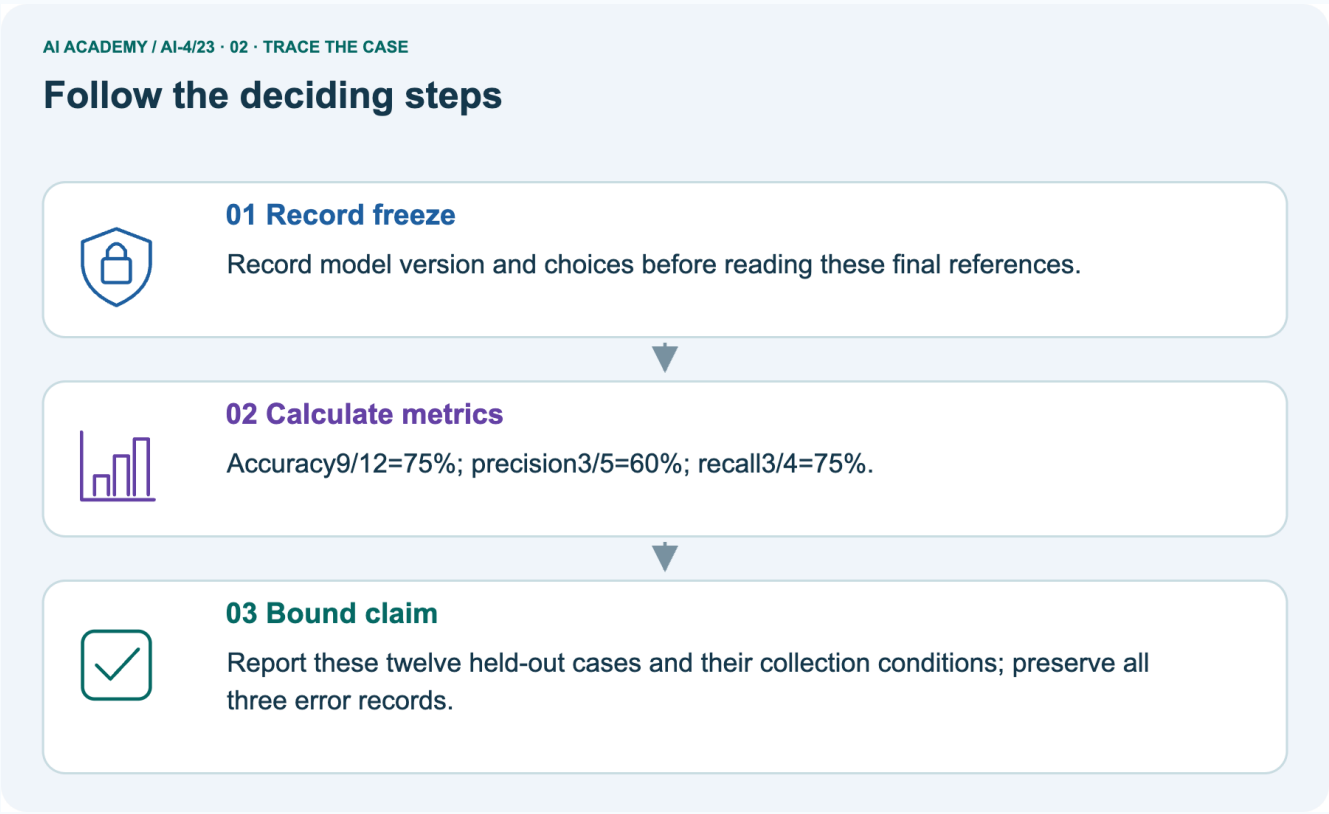
Open the final hold-out only after recording the frozen model choice. Supplied facts: Frozen plant classifier evaluates12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is **claim**ed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



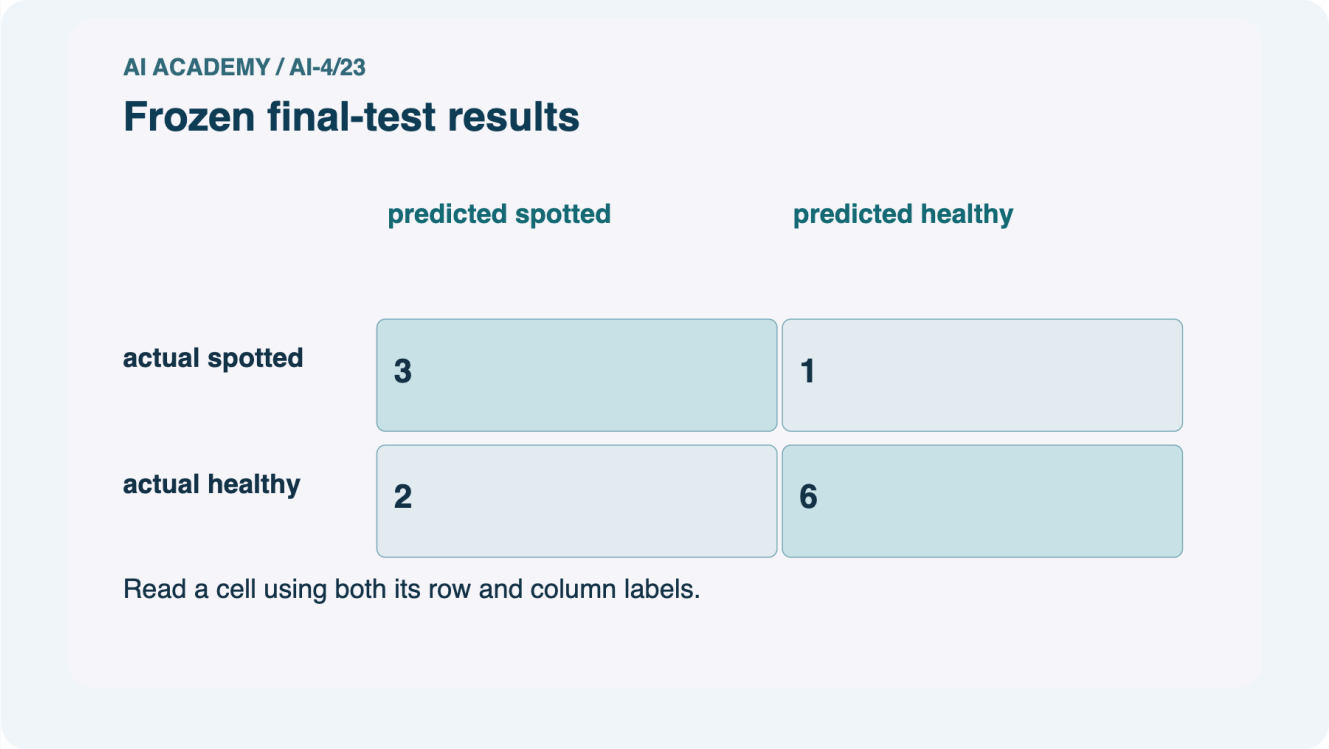
Follow the labelled picture with the matching explanation. Open the vector diagram

Record **freeze**: Record model version and choices before reading these final references. → Calculate metrics: Accuracy9/12=75%; precision3/5=**60%**; recall3/4=75%. → Bound **claim**: Report these twelve held-out cases and their collection conditions; preserve all three error records.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Frozen final-test results



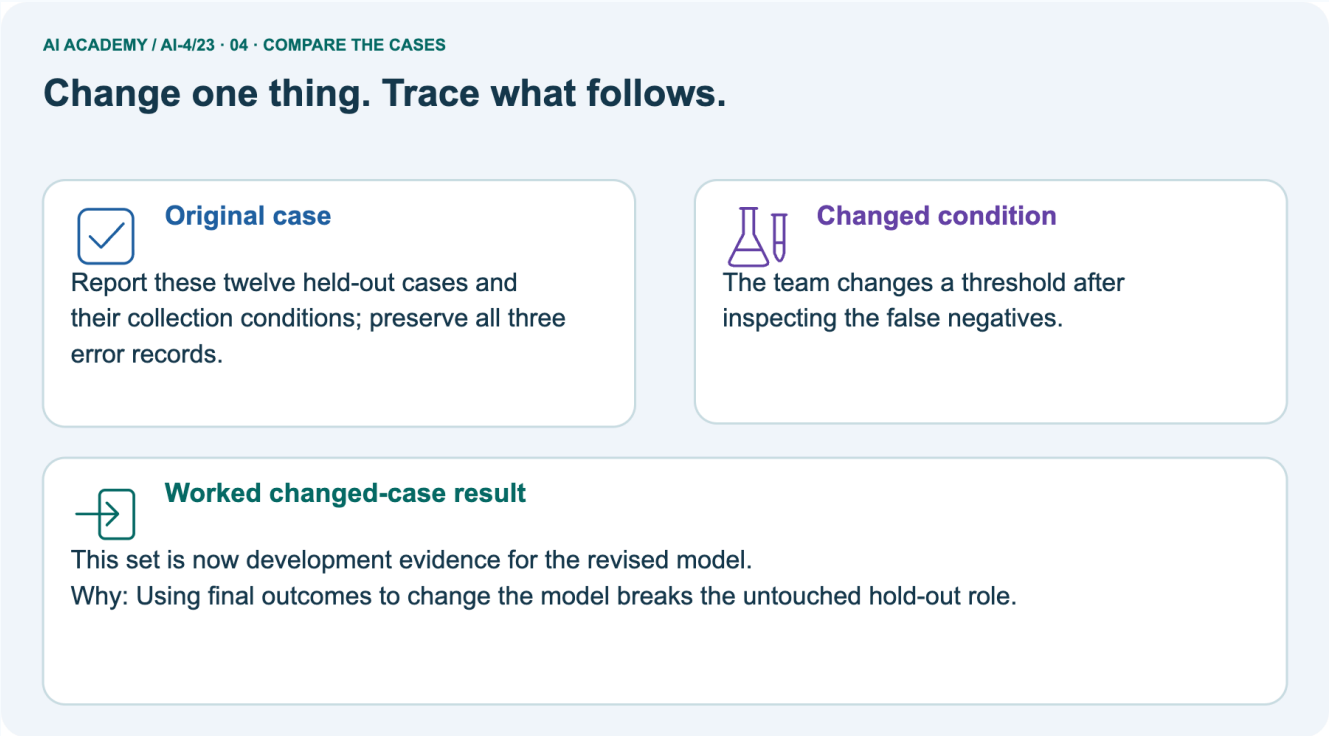
Follow the labelled picture with the matching explanation. Open the vector diagram

Frozen final-test results. actual spotted: predicted spotted = 3, predicted healthy = 1. actual healthy: predicted spotted = 2, predicted healthy = 6.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

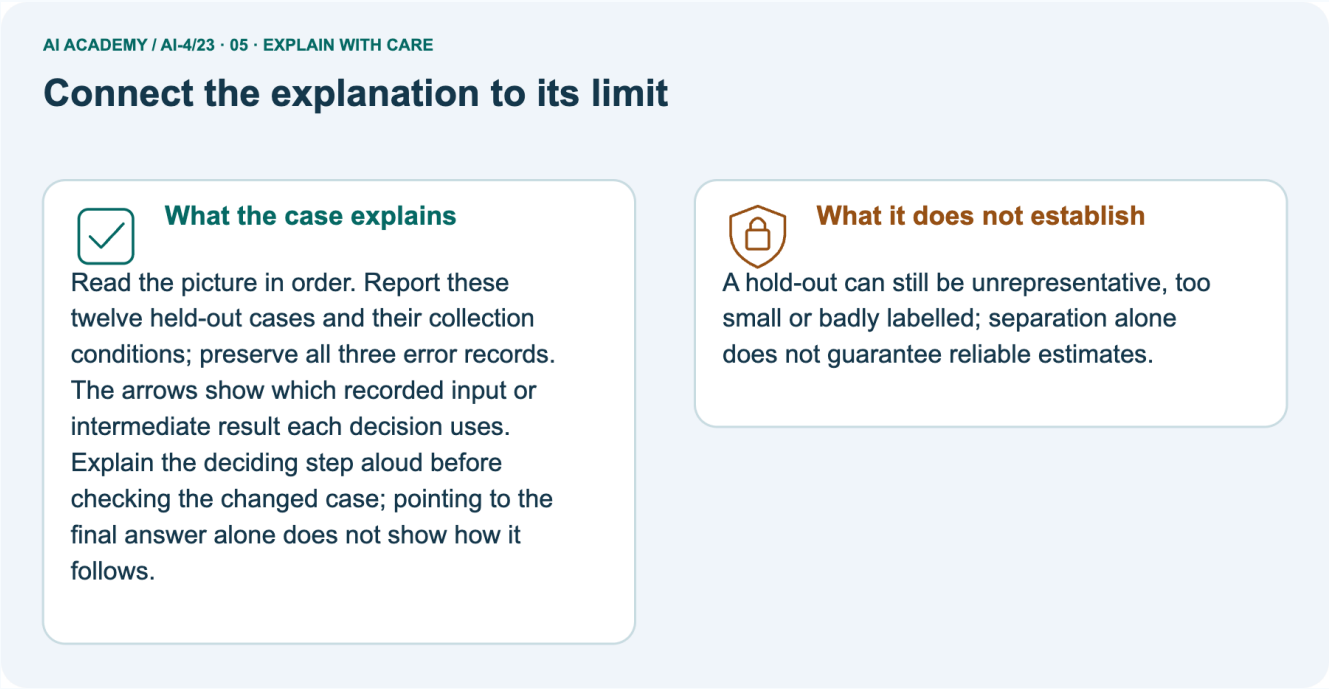
Original case: Report these twelve held-out cases and their collection conditions; preserve all three error records. Changed condition: The team changes a threshold after inspecting the false negatives.

Expected result: This set is now development **evidence** for the revised model. Reason: Using final outcomes to change the model breaks the untouched hold-out role.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

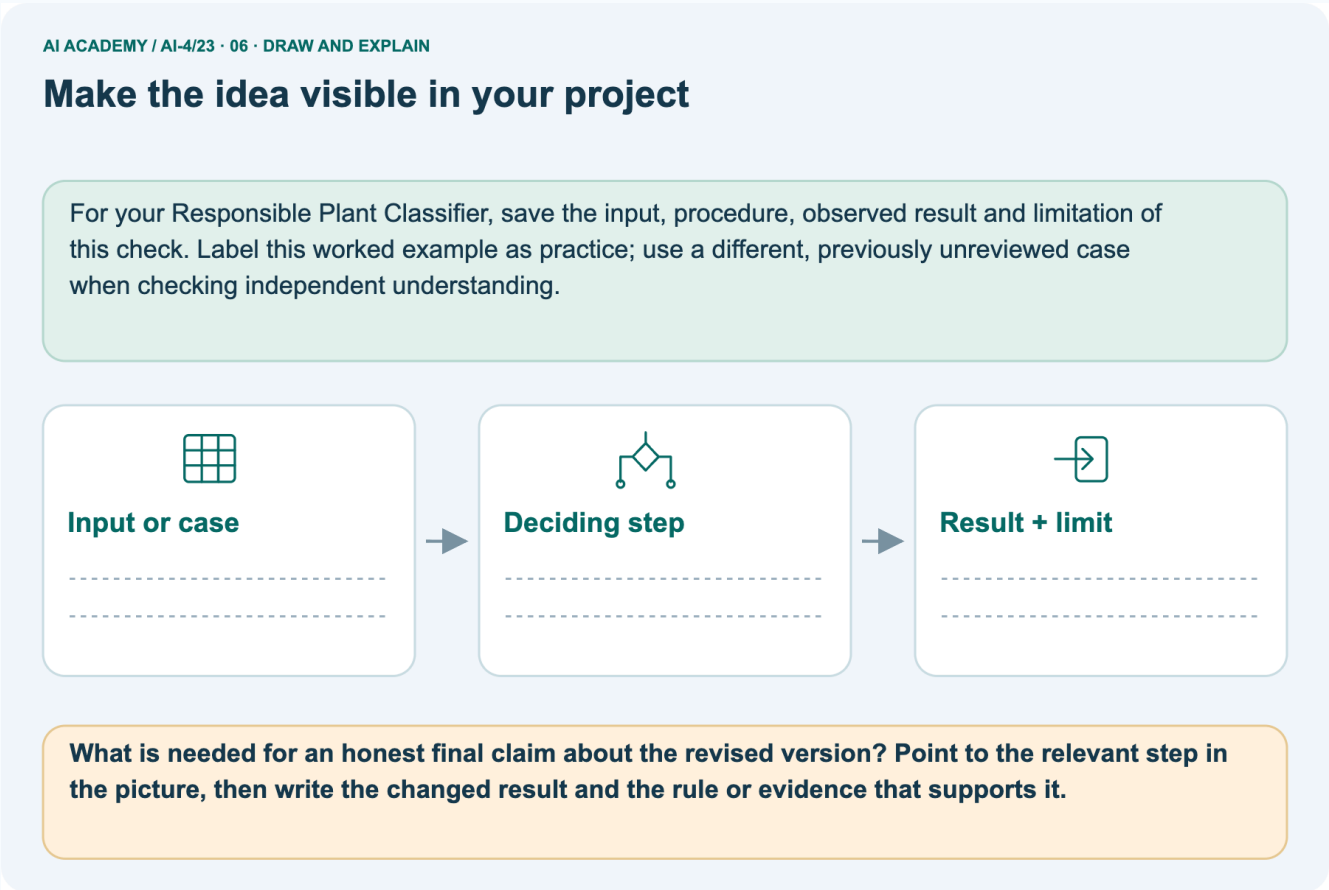
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A hold-out can still be unrepresentative, too small or badly labelled; separation alone does not guarantee reliable estimates.

Explain the case, then point to a limit on the **claim**.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.